PRIMES THAT ARE WDD AND DTUS

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Abstract

DTUS Abstract

1. Introduction

In 1979, P. Erdös [3] proved that there are infinitely many *digitally delicate primes*, a prime that becomes composite when changing any digit to any another digit. The first such prime is 294001, meaning that each number in the set

$$\{d94001, 2d4001, 29d001, 294d01, 2940d1, 29400d : d \in \{0, 1, 2, \ldots, 9\}\}$$

is either 294001 or composite. Since then, the study of such primes has expanded. In 2021, M. Filaseta and J. Southwick [8] extended Erdös' result to also consider the infinitely many leading zeros.

Definition 1 (Widely Digitally Delicate)**.** A prime number is called *widely digitally delicate* if changing any of its base 10 digits, including any of the infinitely many leading zeros, to another base 10 digit results in a composite number.

In the same paper [8], M. Filaseta and J. Southwick showed that there are infinitely many widely digitally delicate primes. While 294001 is digitally delicate, it is not widely digitally delicate as 10294001 is prime. The first prime found to be widely digitally delicate was discovered by J. Grantham [9] and has 4032 digits.

M. Filaseta and the first author [4] refined the methods used in [8] to show that for every positive integer k, there exist k consecutive widely digitally delicate primes. There proof utilized covering systems.

Definition 2 (Covering System)**.** A *covering system* (or *covering*) is a finite set of congruences

$$x \equiv a_1 \pmod{b_1}, \quad x \equiv a_2 \pmod{b_2}, \quad \ldots, \quad x \equiv a_r \pmod{b_r},$$

where $r \in \mathbb{Z}^+$, each $a_i \in \mathbb{Z}$, and each $b_i \in \mathbb{Z}^+$, such that every integer satisfies at least one congruence in the finite set of congruences.

In [4] the authors created 18 coverings to create an arithmetic progression $An+B$ such that A and B are relatively prime and every prime in the arithmetic progression is widely digitally delicate. To conclude their proof, they combined this arithmetic progression with a result due to D. Shiu [16]. See also [1], [2], and [13].

Before discussing the next extension of widely digitally delicate primes, we first consider a few different classes of odd integers. An odd integer k is a *Sierpiński number* if $k \cdot 2^n + 1$ is composite for every positive integer n. In 1960, W. Sierpiński [17] proved that their are infinitely many Sierpiński numbers by showing that every positive integer in the arithmetic progression

$$k = 15511380746462593381 + 36893488147419103230m$$

is a Sierpiński number. The smallest known Sierpiński number is $k = 78557$ and was discovered by J. Selfridge [10, Section B21] using covering systems. In the past few decades various groups have been working on verifying that $k = 78557$ is in fact the smallest Sierpiński number. L. Heim and D. Norris started The Seventeen or Bust project in 2002 to check the remaining 17 values of k less than 78557 that could still be Sierpiński numbers (see [14]). The project showed that 11 of the remaining 17 values of k were not Sierpiński numbers by exhibiting a value for n that makes $k \cdot 2^n + 1$ prime. Then PrimeGrid took over and as of February 2021, the remaining possible Sierpiński numbers less than 78557 are 21181, 22699, 24737, 55459, and 67607.

Around the same time that Sierpiński was looking at numbers of the form $k \cdot 2^n + 1$, Riesel [15] showed that every positive integer in the arithmetic progression $k = 509203 + 11184810m$ is such that $k \cdot 2^n - 1$ is composite for every positive integer n. Such odd integers k are called Riesel numbers and the smallest known Riesel number is $k = 509203$. Similar to the Seventeen or Bust project, the Riesel Sieve Project formed in August 2003 set out to prove that every odd integer less than 509203 is not a Riesel number. PrimeGrid took over in 2021 and as of January 2021, there are only 42 integers less than 509203 left to consider.

A Brier number is an odd integer that is both a Sierpiński number and a Riesel number. Brier (see [4]) proved the existence of such integers by showing that

$$29364695660123543278115025405114452910889$$

is both a Sierpiński number and a Riesel number. The smallest known Brier number is 3316923598096294713661 found by C. Clavier (see [18, A076335]) in December 2013.

In 2023, M. Filaseta and both authors [5] combined the notion of widely digitally delicate primes with Brier numbers. They showed that for any positive integer k, there are k consecutive primes each of which are both widely digitally delicate and a Brier number. The proof consists of finding coprime positive integers A and B such that every prime in the arithmetic progression $An+B$ satisfies the criteria of being widely digitally delicate and a Brier number. With the arithmetic progression, a result due to Shiu [16] proves their result. Shiu showed that in any arithmetic progression with infinitely many primes, there are arbitrarily long sequences of primes. For the arithmetic progression $An+B$ to contain infinitely many primes, we need A and B to be relatively prime. Shiu's result has been strengthened by J. Maynard [12] by showing that in any arithmetic progression $An+B$ for $A>0$, $B \geq 0$, and $n \geq 0$ all integers with A and B relatively prime and fixed, a positive proportion of positive integers ℓ are such that $p_\ell, p_{\ell+1}, \ldots, p_{\ell+k-1}$ are all in the arithmetic progression $An+B$, where p_j denotes the jth prime. Using Maynard's result, M. Filaseta and both authors [5] showed that a positive proportion of the primes are both widely digitally delicate and a Brier number.

Recent work by the second author and R. J. Philpott [11] further extends the notion of widely digitally delicate primes as follows.

Definition 3 (d-Translated Unit Sensitive)**.** For a nonnegative integer d, a prime p is *d-translated unit sensitive* if, after appending a string of length d of zeros to the right of the units digit of p, the resulting number is composite and remains composite when changing the new units digit to any other digit. In the case of $d=0$, we say p is *unit sensitive.*

For example, the first digitally delicate prime, 294001, is not a 5-translated unit sensitive prime as 29400100003 is prime. The second author and Philpott [11] extended this notion finding the first example of a prime that is d-translated unit sensitive for any nonnegative integer d. These primes are called $[0,\infty)$*-translated unit sensitive.* Since the methods used in the construction were similar to that of Filaseta and both authors of this paper, the second author and Philpott [11] found that a positive proportion of primes are $[0,\infty)$-translated unit sensitive.

In this paper, we note that the methods used in [4] produce an arithmetic progression such that every prime in the progression is $[0,\infty)$-translated unit sensitive, widely digitally delicate, and in the case of [5], also a Brier number.

Theorem 1. *The methods used in [4] and [5] to show that there are k-consecutive widely digitally delicate primes for every positive integer k imply that these widely digitally delicate primes are also $[0,\infty)$-translated unit sensitive.*

We also note that Maynard's extension of Shiu's theorem implies that a positive

proportion of the primes are widely digitally delicate, Brier numbers, and $[0, \infty)$-translated unit sensitive primes. The main result in this paper extends the idea of a $[0, \infty)$-translated unit sensitive prime to look at appending a string of nines to the right of the units digit instead of a string of zeros. For ease of notation, we define this as follows.

Definition 4 (Unitarily Unstable)**.** A prime p is *unitarily unstable* if, after appending a string of length k of nines to the right of the units digit of p for any positive integer k, the resulting number is composite and remains composite when changing the new units digit to any other digit.

For example, the first digitally delicate prime, 294001, is not unitarily unstable as appending three nines results in the prime number 294001999. Moreover, appending seven nines and then changing the units digit to a 3 results in the prime number 2940019999993. Our main result is the following.

Theorem 2 (Main Result)**.** *For every positive integer k, there exist k consecutive primes each of which is a widely digitally delicate prime, a Brier number, a $[0, \infty)$-translated unit sensitive prime, and a unitarily unstable prime, with the exception that when appending a string of a multiple of 8704 nines to the right of the units digit where can potentially result in a prime number.*

Our main result uses the techniques in [4] and [5] to create an arithmetic progression $An + B$ such that every prime in the progression is widely digitally delicate, a Brier number, and a unitarily unstable prime. The proof of Theorem 1 implies that the primes in the progression $An + B$ will also be $[0, \infty)$-translated unit sensitive primes. The result due to Shiu will then finish Theorem 2. As with Theorem 1, we also note that Maynard's extension of Shiu's theorem implies that a positive proportion of the primes are widely digitally delicate, Brier numbers, $[0, \infty)$-translated unit sensitive primes, and unitarily unstable primes. We also note that the value $8704 = 17 \cdot 2^9$ can be extended to 17 times a larger power of 2.

2. General Technique

As mentioned in the introduction, to establish Theorem 2 we will construct an arithmetic progression $An + B$ with certain criteria on A and B. First, the positive integers A and B will be fixed where $\gcd(A, B) = 1$. We will require A to be divisible by 10, the importance of which will be explained more in Section 3, and have a prime divisor greater than 10. The choice of A and B will be made such that every integer m of the form $An + B$ satisfies the following criteria:

(i) for $x \in \{-9, \ldots, -1, 1, \ldots, 9\}$, each number in the set

$$\mathcal{A}_x = \mathcal{A}_x(A, B) = \{An + B + x \cdot 10^k : n, k \in \mathbb{Z}^+ \cup \{0\}\}$$

is divisible by at least one prime p dividing A.

(ii) each number in the set

$$\mathcal{B}_S = \mathcal{B}_S(A, B) = \{(An + B) \cdot 2^k + 1 : n, k \in \mathbb{Z}^+ \cup \{0\}\}$$

is divisible by at least one prime p dividing A.

(iii) each number in the set

$$\mathcal{B}_R = \mathcal{B}_R(A, B) = \{(An + B) \cdot 2^k - 1 : n, k \in \mathbb{Z}^+ \cup \{0\}\}$$

is divisible by at least one prime p dividing A.

(iv) for $d \in \{0, \ldots, 9\}$, each number in the set

$$\mathcal{D}_d = \mathcal{D}_d(A, B) = \{(An + B) \cdot 10^\ell + d : n \in \mathbb{Z}^+ \cup \{0\}, \ell \in \mathbb{Z}^+\}$$

is divisible by at least one prime p dividing A.

(v) for $d \in \{1, \ldots, 10\}$, each number in the set

$$\mathcal{U}_d = \mathcal{U}_d(A, B) = \{(An + B + 1) \cdot 10^k - d : n \in \mathbb{Z}^+ \cup \{0\}, k \in \mathbb{Z}^+\}$$

is divisible by at least one prime p dividing either A, except potentially when $k \equiv 0 \pmod{8704}$ and $d = 1$.

Notice that (i) is the criteria for being a widely digitally delicate prime; (ii) and (iii) ensure that m is a Brier number; (iv) ensures that m is $[0, \infty)$-translated unit sensitive; and (v) is the criteria for being unitarily unstable.

Assuming that we can find such a progression where, for $m = An + B$, we have if $m + x \cdot 10^k$, $m \cdot 2^k + 1$, $m \cdot 2^k - 1$, and $m \cdot 10^k + d$ are all not equal to a prime divisor of A, and if $(m + 1) \cdot 10^k - d$ is not equal to a prime divisor of A or B except in the special case $k \equiv 0 \pmod{8704}$ and $d = 1$, then Theorem 2 will follow from Shiu's theorem. Observe that $m \cdot 2^k + 1$, $m \cdot 2^k - 1$, $m \cdot 10^k + d$, and $(m + 1) \cdot 10^k - d$ are each larger than A since B and n are nonnegative, and hence can't be a prime divisor of A. Lemma 1 of [5], with $b = 10$, proves that we can find a subprogression $A_0 n + B_0$ of $An + B$ such that every integer in the subprogression satisfies $A_0 n + B_0 + x \cdot 10^k$ is divisible by a prime p dividing A_0 while not equaling p for each $x \in \{-9, \ldots, -1, 1, \ldots, 9\}$. Thus, we can find a subprogression of $An + B$ such that every prime in the subprogression is widely digitally delicate, a Brier number, $[0, \infty)$-translated unit sensitive, and unitarily unstable.

For each of the five criteria above, we will construct coverings of the integers for k where each congruence in the covering corresponds to a congruence condition on A and B. In the next section we will discuss the construction of the coverings used for (i) and (v). We will not discuss the creation of the coverings for (ii) and (iii) as these

coverings remain unchanged from [5]. Details on why we can use these coverings can be found in Section 4. In Section 4 we also justify why we can use the coverings used for (i) to also satisfy (iv), with a small modification. This justification will provide a proof of Theorem 1.

3. The Coverings for (i) and (v)

As discussed in the previous section, we are interested in finding coprime A and B such that (i) through (v) hold and A is divisible by 10 and a prime greater than 10. This will lead to an arithmetic progression $An + B$ that, combined with Shiu's Theorem, will prove Theorem 2. For each of the criteria (i) through (v) we will construct coverings for k, which correspond to congruence conditions on A and B. In this section we will discuss the coverings used for (i) and (v) and the corresponding congruence conditions on A and B.

The first covering we consider is $k \equiv 0 \pmod 1$, which corresponds to the prime 3. For (i) to hold, we must have every number in the set $\mathcal{A}_x = \{An + B + x \cdot 10^k : n, k \in \mathbb{Z}^+ \cup \{0\}\}$ be composite for each $x \in \{-9, \ldots, -1, 1, \ldots, 9\}$. Looking at the prime 3, we will take

$$A \equiv 0 \pmod 3 \quad \text{and} \quad B \equiv 2 \pmod 3.$$

Since $10^k \equiv 1 \pmod 3$ for any nonnegative integer k, we have $An+B+x\cdot 10^k \equiv 2+x \pmod 3$; therefore, whenever $x \equiv 1 \pmod 3$, each number in the set $\mathcal{A}_x$ is divisible by 3, which by construction is a prime dividing A. This takes care of $\mathcal{A}_x$ for each $x \in \{-8, -5, -2, 1, 4, 7\}$.

Additionally, for (v) to hold, we must have every number in the set $\mathcal{U}_d = \mathcal{U}_d(A, B) = \{(An + B + 1) \cdot 10^k - d : n \in \mathbb{Z}^+ \cup \{0\}, k \in \mathbb{Z}^+\}$ be composite for each $d \in \{1, \ldots, 10\}$ except potentially when $k \equiv 0 \pmod{8704}$ and $d = 1$. Since $B \equiv 2 \pmod 3$ and $A \equiv 0 \pmod 3$, we have $(An + B + 1) \cdot 10^k - d \equiv -d \pmod 3$. Hence, as long as $d \equiv 0 \pmod 3$, each number in the set $\mathcal{U}_d$ is divisible by 3. This takes care of $\mathcal{U}_d$ for each $d \in \{3, 6, 9\}$. Moreover, $\mathcal{U}_d$ is even for $d \in \{2, 4, 6, 8, 10\}$, and divisible by 5 for $d \in \{5, 10\}$. Notice that this is one instance of why we want A to be divisible by 10.

The next covering we consider is also $k \equiv 0 \pmod 1$, this time corresponding to the prime 7. Looking at (v), if we take

$$A \equiv 0 \pmod 7 \quad \text{and} \quad B \equiv 6 \pmod 7,$$

then $(An + B + 1) \cdot 10^k - d \equiv -d \pmod 7$. Therefore, whenever $d \equiv 0 \pmod 7$, each number in the set $\mathcal{U}_d$ is divisible by 7, which takes care of $d = 7$. We now need coverings for $x \in \{-9, -7, -6, -4, -3, -1, 2, 3, 5, 6, 8, 9\}$ for $\mathcal{A}_x$ in criteria (i), and $d = 1$ for $\mathcal{U}_d$ in criteria (v).

Before proceeding to a more complicated covering, we note that we will be restructuring the coverings used in [4] (and therefore [5]) for criteria (i) to ensure the primes in the arithmetic progression $An+B$ are widely digitally delicate. In [4] the authors took $B \equiv 1 \pmod 3$ and $B \equiv 1 \pmod 7$; whereas we are taking $B \equiv 2 \pmod 3$ and $B \equiv 6 \pmod 7$. By restructuring the coverings to accommodate this change in B, we reduce the total number of coverings needed for criteria (v) from four, namely $d \in \{1, 3, 7, 9\}$, to only one, namely $d = 1$.

We now discuss precisely how a covering for a specific x or d value will be used. Towards criteria (i), let $k \equiv a_i \pmod{b_i}$ be a congruence in the covering for a specific x value where b_i is the order of 10 modulo the prime p_i. We will impose the congruence conditions

$$A \equiv 0 \pmod{p_i} \qquad \text{and} \qquad B \equiv -x \cdot 10^{a_i} \pmod{p_i}.$$

Note that with these congruence conditions on A and B, the value $An+B+x \cdot 10^k$ will be divisible by p_i for the specified x value when $k \equiv a_i \pmod{b_i}$. Also note that $-x \cdot 10^{a_i}$ will never be divisible by p_i as 7 will be the only prime less than 10 used, and 7 will not be used in the coverings for x when x is divisible by 7.

Similarly, for criteria (v) let $k \equiv a_i \pmod{b_i}$ be a congruence in the covering for $d = 1$ where b_i is the order of 10 modulo the prime p_i. We will impose the congruence conditions

$$A \equiv 0 \pmod{p_i} \qquad \text{and} \qquad B \equiv 10^{p_i - a_i - 1} - 1 \pmod{p_i}$$

where the exponent on 10 comes from using Fermat's Little Theorem to find the multiplicative inverse of 10^{a_i} modulo p_i. With these congruence conditions on A and B, the value $(An+B+1) \cdot 10^k - 1$ will be divisible by p_i when $k \equiv a_i \pmod{b_i}$. It is important to note here that in order for A and B to be coprime, we must never let $B \equiv 0 \pmod{p_i}$. This happens precisely when $-a_i$ is a multiple of b_i, the order of 10 modulo p_i, which corresponds to $a_i \equiv 0 \pmod{b_i}$. In the covering used for $d = 1$, the only time this happens is when $k \equiv 0 \pmod{8704}$ corresponding to the prime 2350081. For this one congruence we will impose no congruence condition on B to keep A and B relatively prime. As mentioned below the statement of Theorem 2, we can replace the congruence $k \equiv 0 \pmod{8704}$ with the congruences $k \equiv 0 \pmod{8704 \cdot 2}$ and $k \equiv 8704 \pmod{8704 \cdot 2}$. We can repeat this process of replacing the congruence $k \equiv 0 \pmod{b_i}$ to make the modulus b_i as large as we want.

To ensure such a B exists, we will limit the number of times the modulus b_i appears in the covering by the number of unique primes we can find such that 10 has order b_i modulo p_i. Each prime divisor p of $\Phi_b(10)$, the bth cyclotomic polynomial, will either be the largest prime divisor of b or will satisfy 10 has order b modulo p (see [4], [5], and [6]). Thus, we can use the modulus b_i a number of times equal to the number of prime factors p_i of $\Phi_{b_i}(10)$ that are coprime to b_i.

This is what complicates creating the coverings, we can not, in general, use a prime more than once which greatly limits the number of times we can use small moduli. The number of congruences used in the coverings for criteria (i) and (v) can be found in Table 1.

x	# cong.
-9	69
-8	1
-7	388
-6	408
-5	1
-4	143
-3	29
-2	1
-1	882

x	# cong.
1	1
2	538
3	4
4	1
5	226
6	59
7	1
8	4
9	14

d	# cong.
1	560
2	1
3	1
4	1
5	1

d	# cong.
6	1
7	1
8	1
9	1
10	1

Table 1: Number of congruences for each covering for criteria (i) and (v)

To illustrate the use of the coverings for criteria (i), consider the covering used for $x = 3$ in Table **??**, which contains four congruences. These congruences are

$$k \equiv 0 \pmod 2, \quad k \equiv 0 \pmod 3, \quad k \equiv 5 \pmod 6, \quad \text{and} \quad k \equiv 1 \pmod 6.$$

Each congruence corresponds to a different prime p_i with 10 having order modulo p_i equal to the modulus b_i. We chose the prime 11 with 10 of order 2, the prime 37 with 10 of order 3, the prime 7 with 10 of order 6 for the third congruence, and the prime 13 with 10 of order 6 for the last congruence. For each of the congruences, set $A \equiv 0 \pmod{p_i}$ and $B \equiv -3 \cdot 10^{b_i} \pmod{p_i}$. That is, we set $A \equiv 0 \pmod{11}$, $A \equiv 0 \pmod{37}$, $A \equiv 0 \pmod 7$, and $A \equiv 0 \pmod{13}$; and $B \equiv 8 \pmod{11}$, $B \equiv 34 \pmod{37}$, $B \equiv 6 \pmod 7$, and $B \equiv 9 \pmod{13}$. Thus, the value $An + B + 3 \cdot 10^k$ will be divisible by 11, 37, 7, or 13 depending on the value of k.

We note that there are quite a few choices to be made when constructing the coverings. Two noteworthy decisions are the covering for $x = 3$ starts with the congruence $k \equiv 0 \pmod 2$ instead of $k \equiv 1 \pmod 2$, and we use the prime 7 for the congruence $k \equiv 5 \pmod 6$ instead of the congruence $k \equiv 1 \pmod 6$. While these choices are arbitrary and we could change these decisions, these decisions were chosen specifically to help make later coverings easier to construct.

To illustrate the importance of these decisions, consider the prime 7. We previously said that we would only use the modulus b_i a number of times equal to the number of unique primes we can find such that 10 has order b_i modulo p_i. This is done to ensure that a B exists that satisfies all of the congruence conditions. Based on Table **??**, 10 has order 7 modulo only two primes, 7 and 13. We already used

the prime 7 above to take care of $d = 7$ in criteria (v), yet we use it again in the covering for $x = 3$. We are able to use a prime in multiple locations, thus repeating the modulus b_i more times than expected, as long as the congruence condition on B remains the same. When dealing with $d = 7$ at the beginning of this section, we imposed the congruence condition $B \equiv 6 \pmod 7$, which is the same congruence condition on B imposed by the congruence $k \equiv 5 \pmod 6$ in the covering for $x = 3$. Most of the decisions made on the construction of the coverings are made to maximize the number of times we are able to reuse a prime corresponding to a small modulus. Table 2 gives a list of all the primes that are used in multiple coverings for conditions (i) and (v). The only prime to be reused in the coverings for conditions (ii) and (ii) is 5. The function $\rho(b, p)$ will be explained in the next section.

p	congruence on B	coverings used	$\rho(b,p)$
7	$6 \pmod 7$	$x = -9, -6, -4, -3, -1, 2, 3, 5, 9$	1
11	$8 \pmod{11}$	$x = -3, 3, 8$	1
13	$9 \pmod{13}$	$x = -9, -4, -3, -1, 3, 9, d = 1$	2
17	$15 \pmod{17}$	$x = -7, -6, 2, 4, d = 1$	1
19	$3 \pmod{19}$	$x = -6, -1,\ d = 1$	1
23	$22 \pmod{23}$	$x = -7, 5$	1
29	$22 \pmod{29}$	$x = -7, 2, 6, d = 1$	1
31	$29 \pmod{31}$	$x = -6, 2, 9$	1
37	$34 \pmod{37}$	$x = -7, 3$	1
41	$32 \pmod{41}$	$x = 5, 9$	1
43	$12 \pmod{43}$	$x = 6, d = 1$	1
53	$37 \pmod{53}$	$x = -7, -6$	1
61	$41 \pmod{61}$	$x = -9, 9, d = 1$	1
71	$25 \pmod{71}$	$x = -9, -6$	1
73	$65 \pmod{73}$	$x = -7, -3, 8$	1
79	$47 \pmod{79}$	$x = -7, -6$	2
89	$78 \pmod{89}$	$x = 5, d = 1$	1
157	$87 \pmod{157}$	$x = -7, -6$	3
211	$171 \pmod{211}$	$x = -9, 9$	1
241	$83 \pmod{241}$	$x = -9, 9$	2
547	$11 \pmod{547}$	$x = -6, 2$	1
2161	$1399 \pmod{2161}$	$x = -9, 9$	3
9091	$900 \pmod{9091}$	$x = -9, 9$	1
2906161	$2615544 \pmod{2906161}$	$x = 9, d = 1$	2
4188901	$795510 \pmod{4188901}$	$x = 9, 9$	2

Table 2: Primes used for more than one covering

While most of the primes used in our coverings can be found in [4] and [5], we

had to find more primes because of the additional covering for $d = 1$. Using Magma V2.25-7 we were able to find the remaining prime factors that we needed to complete our coverings. For data transparency, we have made available the list of these new primes associated with each modulus b available through [**?**]. Magma was not always able to fully factor $\Phi_{b_i}(10)$ in a reasonable time. When this happened, we used the remaining unfactored part by verifying that is is composite, relatively prime to the factored part of $\Phi_{b_i}(10)$ and b_i, and not a prime power. When this happened we were able to say that the unfactored part has at least two more unique prime factors, thus allowing us to use the modulus b_i twice more. Using the unfactored part, we were able to get a lower bound for the number of distinct primes of a given order b, denoted $L(b)$. A complete list of the moduli used, b, with a lower bound on the prime divisors of $\Phi_b(10)$, corresponding to a lower bound on the number of primes for which 10 has order b, is found in Table **??**.

Tables **??**-**??** give the congruences $k \equiv a_i \pmod{b_i}$ that form the coverings for $x \in \{-9, -7, -6, -4, -3, -1, 2, 3, 5, 6, 8, 9\}$ and $d = 1$.

4. Other Coverings and a Proof of Theorem 1

With the coverings created as described in Section 3, we have an arithmetic progression $An+B$ with such that criteria (i) and (v) hold from the beginning of Section 2. That is, every prime in the arithmetic progression $An+B$ is widely digitally delicate and unitarily unstable, except potentially when appending a number of nines to the right of the units digit that is a multiple of 8704. We now discuss criteria (ii), (iii), and (iv).

Towards criteria (ii) and (iii), we impose the congruence conditions $A \equiv 0 \pmod 2$ and $B \equiv 1 \pmod 2$. This ensures that every number in the arithmetic progression $An + B$ is odd. Now criteria (ii) corresponds to ensuring that every integer in the arithmetic progression $An + B$ is a Sierpiński number, and criteria (iii) ensures they are Riesel numbers. The two coverings used for criteria (ii) and (iii) can be found in [5]. These two coverings remain unchanged; for the sake of brevity we will not recreate them here. Instead we make a few observations about their construction and why we can use them in our case.

The construction of the coverings for criteria (ii) and (iii) found in [5] was done in much the same way as the coverings for criteria (i) and (v) found here. The key difference is these coverings are for the exponent on 2, not on 10. That means when finding primes associated with the modulus b in the coverings, we no longer find prime factors of $\Phi_b(10)$, but instead find prime factors of $\Phi_b(2)$. The covering used for criteria (ii) containing 447 congruences can be found in Table 5 of [5], and the covering used for criteria (ii) containing 459 congruences can be found in Table 6 of [5]. The only prime used in both coverings is the prime 5.

In order to use these two coverings as created in [5], we need to verify two things. First, we verify that none of the primes used in Tables **??**-**??** for criteria (i) and (v) are used in the two coverings found in Tables 5 and 6 of [5]. Second, we verify that each composite number used in any of the coverings is coprime to each other composite number used, and coprime to each prime used. A quick computation confirms both of these requirements.

With criteria (i), (ii), (iii), and (v) satisfied by the coverings found in Tables **??**-**??** and Tables 5 and 6 in [5], we turn our attention to criteria (iv) which guarantees that each prime in the arithmetic progression $An + B$ is $[0, \infty)-$translated unit sensitive. As stated in Theorem 1, the coverings used in criteria (i) satisfy criteria (iv) as well. To show this, we now give a proof of Theorem 1.

Proof of Theorem 1. Assume that we have an arithmetic progression $An + B$ with $\gcd(A, B) = 1$ that satisfies the conditions of criteria (i) by creating the 18 coverings described in Section 3. Note that if we append k zeros to the right of the units digit of $An+B$, then the resulting number is divisible by 10 and hence composite. So we need not consider $d = 0$ in criteria (iv). By construction, each number of the form $An + B + x \cdot 10^k$ is composite for $x \in \{1, \ldots, 9\}$ for any nonnegative integers k and n. The construction of A and B is such that for a fixed x, we have the covering $\mathcal{K}_x := \{k \equiv k_i \pmod{b_i} : 1 \leq i \leq j_x\}$ where b_i is the order of 10 modulo p_i and j_x is the number of congruences in the covering used for x. For each congruence in $\mathcal{K}_x$, we take $B \equiv -x \cdot 10^{k_i} \pmod{p_i}$ and $A \equiv 0 \pmod{p_i}$.

For these primes to be $[0, \infty)$-translated unit sensitive, we must also have that every number of the form $(An + B) \cdot 10^\ell + d$ is composite for $d \in \{1, \ldots, 9\}$ and ℓ and n nonnegative integers. For a fixed d, consider the covering used for $\mathcal{K}_x$ for $x = d$. Since $\mathcal{K}_x$ is a covering, every integer, both positive and negative, satisfies one of the congruences in $\mathcal{K}_x$. Thus, taking the negative residue of each congruence will also result in a covering, namely $-\mathcal{K}_x := \{\ell \equiv -k_i \pmod{b_i} : 1 \leq i \leq j_x\}$.

For each congruence in $-\mathcal{K}_x$, from criteria (i) we still have $A \equiv 0 \pmod{p_i}$ and $B \equiv -x \cdot 10^{k_i} \pmod{p_i}$ where the prime p_i was assigned to the congruence $k \equiv k_i \pmod{b_i}$ in $\mathcal{K}_x$. Using the same primes now assigned to $\ell \equiv -k_i \pmod{b_i}$ in $-\mathcal{K}_x$, we impose no additional criteria on A or B. For each congruence in $-\mathcal{K}_x$, for $d = x$ it follows that

$$(An + B) \cdot 10^\ell + d \equiv -x \cdot 10^{k_i} \cdot 10^{-k_i} + d \equiv 0 \pmod{p_i}.$$

Thus, taking the negative residue of each congruence in each of the coverings used for $x \in \{1, \ldots, 9\}$ in criteria (i) creates new coverings for ℓ such that the progression $An + B$ also satisfies criteria (iv). Doing so does not impose more congruence conditions on A or B and Theorem 1 follows. $\square$

5. Verifying the Coverings

With the coverings described we now explain how we verified each is actually a covering of the integers. Let

$$\mathcal{C} = \{k \equiv a_1 \pmod{b_1}, \ldots, k \equiv a_m \pmod{b_m}\}$$

be a set of congruences. The most direct way to verify that $\mathcal{C}$ is a covering is to check that each integer in $[0, \ell)$ satisfies at least one of the congruences where $\ell = \mathrm{lcm}(b_i)$. A computer can easily perform this calculation for small ℓ; however, if ℓ is large then this computation can take an unreasonable amount of time. Table 3 gives ℓ for each of the coverings used, its table number in the appendix, and the maximum prime that divides ℓ.

x	Table	ℓ	max p
-9	5	14636160	11
-7	6	360	5
-6	7	8	2
-4	8	5040	7
-3	9	216	3
-1	9	216	3
2	9	216	3
3	9	216	3

x	Table	ℓ	max p
5	11	11690842363200	31
6	12	36054018000	29
8	13	281080800	13
9	14	16	2
d	Table	ℓ	max p
1	16	12738408883200	19

Table 3: Least common multiple of the moduli for the coverings in each table

To verify $\mathcal{C}$ is a covering when ℓ is large, we ensure that $\mathcal{C}$
CONTINUE HERE WITH CPU. INCLUDE $\rho(b, p)$.

Consider the covering used for $d = 0$. We know that the congruences

$$k \equiv 0 \pmod 3, \qquad k \equiv 1 \pmod 3, \qquad \text{and} \qquad k \equiv 2 \pmod 3$$

form a covering. The covering for $d = 0$ was constructed by considering each of these three congruences separately. To verify this covering, we can instead verify that each of these three congruences is properly covered. When considering $k \equiv 0 \pmod 3$, we used the congruence $k \equiv 0 \pmod 6$ with the prime 7 in Table 2, and constructed $k \equiv 3 \pmod 6$ by looking at moduli that were divisible by 31. To break it down even further, one can check that the set of congruences

$$A := \{k \equiv a_i \pmod{186} : a_i = 6j + 3 \text{ for } j \in \{0, \ldots, 30\}\}$$

covers $k \equiv 3 \pmod 6$. Thus to verify $k \equiv 0 \pmod 3$ is covered, one needs to verify that each of the congruences in A is covered. Some of these congruences are covered by a single congruence, while some take up to 21 congruences. The largest of these collections of congruences has 21 congruences, with least common multiple of the moduli being 12648. Checking each collection of congruences that correspond to $k \equiv 1 \pmod 3$ and $k \equiv 2 \pmod 3$ individually also reduces ℓ to 89812800 and 272160, respectively.

A similar method can be used to verify the covering for $d = 3$ by looking at each of the residue classes modulo 5 separately; for $d = 4$ one can look at the residue classes modulo 13; and for $d = 9$ one can look at the residue classes for 2.

To finish verification of the coverings found in the appendix and made available in [**?**], the following checks should be and were completed. For each congruence $k \equiv a \pmod b$, the modulus b corresponds to a prime p with 10 of order b. To verify this, we checked that p divides $\Phi_b(10)$ and is coprime to b. Furthermore, throughout all coverings, no prime is repeated unless it appears in Table 2. For the primes appearing in Table 2, we verified that the congruence condition on M, found either by $M \equiv -x \cdot 10^b$ or $M \equiv -1 - (d-1) \cdot 10^b$, depending on if the covering is for a value of x or d, is that which appears in Table 2. For each composite factor of $\Phi_b(10)$ used, we verified that the composite number was used at most twice and was not a prime power. Then we made sure the composite number is coprime to each other factor of $\Phi_b(10)$ and b. These verifications guarantee that we get two unused and unique primes from the composite number.

With the verifications of the coverings completed, we have congruence conditions on n and M that guarantee that

$$N = \frac{10^n - 1}{9} + M$$

remains composite when inserting, replacing, or deleting a digit in the leading zeros or the leading ones. To finish the proof of Theorem 2, we need only consider the right-most digits corresponding to M.

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Appendix

To aid in the verification of the computations in this paper, all the data in this appendix can be found in [**?**] as lists suitable for computations. Table **??** gives a lower bound $L(b)$ on the number of distinct primes dividing $\Phi_b(10)$ that are relatively prime to b. Note that the $(*)$ in Table **??** means that at least one of the primes associated with the modulus b appears in Table 2 and appears in more than one covering.

The remaining tables, Tables **??**-**??**, give the congruences $k \equiv a \pmod{b}$ that form the covering systems obtained for $x \in \{-9, -7, -6, -4, -3, -1, 2, 3, 5, 6, 8, 9\}$ pertaining to criteria (i) and $d = 1$ for criteria (v). Recall that for a fixed modulus b, the prime associated with a given congruence class unless the prime appears in Table 2. We simply put a number in the second column to refer to a prime with 10 of order b, and a different number the next time the modulus b appears to indicate a different prime is used. For these tables, an $(*)$ is used when the prime appears in Table 2 and appears in more than one covering. For these congruences, the choice of prime does matter as the congruence condition on B is fixed because the prime appears in Table 2.